

HCMUT Collaboration: Cross-Lingual Opinion Strength Study

3-Month Preparation Roadmap

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The Core Question

Core question: When the same person discusses the same topic in Japanese, English, and Vietnamese, how do their opinion strength and argumentative consistency change?

Approach:

- **Live API** (Gemini, voice-dialogue AI) for adaptive interviewing
- Within-subject design across 3 language conditions

Why This Study?

Three Gaps in Prior Work

1. Limitations of existing methods

- Only static translated questionnaires
- No flexible Q&A possible

2. Zero literature on FLE in Vietnamese speakers

- The Foreign Language Effect has never been tested with VN speakers

3. No prior JA–EN–VI trilingual comparison

Live API enables response-contingent follow-up → opinion strength can be extracted from naturalistic dialogue.

Two Core Metrics

Definitions and Measurement

Metric	Definition	How to Measure
Opinion Strength	Certainty, assertiveness, resistance to counter-arguments	Hedge-word frequency · prosodic features (eGeMAPS) · stance retention after rebuttal
Argumentative Consistency	Stance and premises remain stable across languages	LaBSE multilingual embedding cosine similarity · binary stance agreement rate

Participants, Procedure, Topics

Participants:

- JA–EN bilinguals: NUT students
- JA–EN–VI trilinguals: HCMUT students

Procedure:

- Adaptive Live API interview on the same topic in each language
- Language presentation order is counterbalanced

Design:

- Within-subject (same individual across all language conditions)

Topics:

- Educational and social topics relevant to multicultural co-learning

Why This Matters in Practice

Before multicultural co-learning discussions, run a pre-discussion interview to obtain each student's cross-lingual opinion profile → use it as input for optimal group formation (proof-of-concept).

Goal:

- Combine students with diverse opinion strengths and argumentation styles
- Propose group compositions that improve discussion quality

Target Venues

Venue	Character	Acceptance	APC
IEEE Access (first choice)	Applied focus, technical soundness	~27–28%	\$2,160
IEEE Trans. Affective Computing	Computational study of emotion/certainty/voice	Stricter	Higher

Three essentials for IEEE Access acceptance:

1. Reproducible system implementation (code release)
2. Novel multilingual dataset (audio + transcript + annotation)
3. Metric validity verification

Month 1 (June): Literature Reading

Must-Read Papers

① Foreign Language Effect — Foundations

- Costa et al. (2014). *Your morals depend on language*. PLOS ONE
- Keysar, Hayakawa & An (2012). *The Foreign-Language Effect*. Psych. Sci.
- Circi et al. (2021). *Meta-analysis of FLE in decision-making*. PB&R

② Cultural Frame Switching

- Ramírez-Esparza et al. (2006). *Do bilinguals have two personalities?* JRP

③ Voice & Certainty Quantification

- Eyben et al. (2015). *eGeMAPS*. IEEE T. Affective Computing

④ Adaptive AI Interviewing

- Wuttke et al. (2024). *AI Conversational Interviewing*. arXiv:2410.01824

⑤ Group Formation (Application Background)

- Minimum Entropy Collaborative Groupings (2023). PLOS One

Month 1: Tasks

Concrete Deliverables

- Each member reads an assigned paper and produces a **1-page A4 summary** (JA or EN)
- **Collaborative glossary:** confirm Opinion Strength · Argumentative Consistency · FLE · CFS · within-subject in JA / EN / VI
 - Intent: avoid translation drift; ensure consistent terminology across participant briefings, protocols, and the paper
- **Run a Gemini Live API demo** (Hello-World level)
- **Think of 3 candidate discussion topics** suitable for multicultural co-learning

Month 2 (July): System Design

Additional Reading & Tasks

Additional papers:

- Stab & Gurevych (2017). *Parsing Argumentation Structures*. Comp. Linguistics
- Qi et al. (2023). *Cross-Lingual Consistency of Factual Knowledge*. arXiv:2310.10378
- Vietnamese politeness studies (face, indirect expression)

Tasks:

- **Interview protocol design:** follow-up logic, rebuttal timing, language-switch procedure
- **Live API implementation sketch:** WebSocket, VAD, session management, logging
- **Hedge-word correspondence table** (JA: ～と思います / EN: I think / VI: tôi nghĩ, etc.)
- **IRB preparation:** voice-recording / personal-information consent drafts

Month 3 (August): Pilot & Validity

Pilot Run & Validity Check

Additional reading:

- Bilingualism: Language and Cognition. *Resisting persuasion in cases of ideological conflict*. (basis for rebuttal-follow-up design)

Tasks:

- **Pilot run:** 5–10 JA–EN bilinguals (NUT side) through the full pipeline
- **Human annotation:** agreement between automatic and human ratings (redesign metrics if Cohen's $\kappa < 0.4$)
- **Vietnamese ASR accuracy pre-measurement:** measure Gemini Live API VN recognition rate
- **Complete data pipeline:** audio → Whisper → feature extraction → scoring → visualization
- **HCMUT briefing slides** (English & Vietnamese) finalized before departure

Tentative Roles

NUT × HCMUT Division of Labor

Role	NUT side	HCMUT side
System implementation	Eita	—
JA / EN protocol design	NUT students	—
VN protocol & translation review	—	HCMUT students
Data annotation (VN)	—	HCMUT students
Statistical analysis	Eita	—
Paper writing	Kumoi (lead)	Co-author

Key Risks & Mitigations

Four Risks to Watch

① Vietnamese ASR accuracy

- Live API quality is highest in English → measure VN rate in Month 3, log error rate as a control variable

② FLE effect-size variability

- $g = 0.09\text{--}0.40$ across prior studies → design with null-result publishability in mind

③ Ethics review (IRB)

- IRB · informed consent · anonymization · Google-API audio-transmission consent in both JP and VN

④ Live API session limit

- ~15-min session expiry → use split-session design for long interviews

Next Steps

Team Collaboration Plan

June (Pre-investigation phase)

- Eita's and Hieu's teams hold joint reading sessions on related papers to deepen understanding
- Both teams discuss and exchange ideas internally via **Slack**
- **Eita & Kei** support Khai and Hieu's research progress and provide tasks for the joint collaboration

From July (Regular reporting phase)

- Propose bi-weekly meetings with Prof. Tho
- The two teams will report progress to the professor **every 2 weeks** from early July

Thank You for Listening!

ご清聴ありがとうございました

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Questions and comments welcome



Appendix: Glossary 1

Study Design & Ethics

Counterbalance

Systematically varying the presentation order of multiple conditions (3 languages here) across participants. Controls for order effects (fatigue, practice).

Within-subject design

A design where the same individual participates in all conditions. Removes individual-difference noise and achieves statistical power with fewer participants.

Protocol

A standardized procedure document for the interview. Greeting → question order → closing, written out so any interviewer can run the session identically.

Informed consent

An ethics procedure: fully explain the study's purpose, procedures, risks, and right to withdraw, and obtain written agreement from the participant.

Appendix: Glossary 2-1

Measurement, Statistics & Tech

Human annotation

Researchers manually assigning evaluation values to data. Used to validate the automatic metric.

Cohen's κ (Kappa)

A statistical measure of agreement between two raters (0 to 1). Below 0.4 indicates insufficient agreement (Landis & Koch, 1977).

eGeMAPS

A standard 88-feature set for extracting certainty, emotion, and speaking-style features from voice (Eyben et al., 2015). Used as the prosodic indicator for Opinion Strength.

LaBSE

Google's multilingual sentence-embedding model. Enables semantic similarity computation across languages. Used for Argumentative Consistency.

Measurement, Statistics & Tech

ASR (Automatic Speech Recognition)

Automatic speech recognition. The process by which Live API internally converts voice to text.

null result

A result where the predicted effect is not observed. This study is designed so that "opinions don't change across languages" is also a publishable, meaningful finding.